
Errata for:

**“Probing the Universe: A Geometrical View for
Observers of Spacetime Physics”**

Robin W. Tucker and Timothy J. Walton (Springer: 2026)



A list of typographical errors and correction for the book will be kept here. Corrections are emphasised in **red** text.

Please email TWalton@lancashire.ac.uk if you find any typographical errors not already listed.

page 16: The third displayed equation on this page should read:

$$\mathbf{v}_P(t) = -R_0 \sin(\Phi(t)) \dot{\Phi}(t) \mathbf{i}_1 + R_0 \cos(\Phi(t)) \dot{\Phi}(t) \mathbf{i}_2$$

page 74: The displayed equation below equation (5.36) should read:

$$\begin{aligned} 0 \diamond \xi &= \xi \\ \tau_1 \diamond (\tau_2 \diamond \xi) &= (\tau_1 \boxtimes \tau_2) \diamond \xi \end{aligned}$$

page 177: The line immediately below the first displayed equation should read:

“In the $\mathbf{g}_{\mathbb{R}^3}$ -orthonormal cobasis for. . .”

page 248: The line immediately below the penultimate displayed equation on this page should read:

“Since the $\mathbf{g}$ -trace of $\mathbf{g}$ is $\text{Tr}^{(\mathbf{g})}(\mathbf{g}) = 4$, taking. . .”

page 285: The line directly above the only displayed equation on this page should read:

“Since $\Pi_Q^\perp(W') = \mathbf{W}$, both the spacelike. . .”

page 290: The sentence directly preceding equation (15.45) should read:

“Recalling $W \equiv \Pi_Q^\perp(W')$ where Q is geodesic with $\mathbf{g}(Q, Q) = -L_0^2$ and W' is Lie-parallel with respect to Q , the vector fields Q, W **satisfy** the conditions. . .”

page 376: The last displayed equation on this page should read:

$$\begin{aligned}\mathcal{G}_c &= \mathcal{R} \star e_c - 2(i_a P_c) \star e^a + \mathbf{2}(i_c i_a i_b \mathfrak{D} T^b) \star e^a \\ &= \mathcal{R} \star e_c - 2 \star P_c - 2i_a(i_c i_b \mathfrak{D} T^b) \star e^a\end{aligned}$$

page 380: The line immediately below the first displayed equation should read:

“...is a type-preserving *derivation* on the algebra of exterior forms on $\mathcal{M}$ (**see appendix H for further details**) that commutes with the exterior derivative...”

There is also a typographical error here: the 8th line below equation (20.8) should read

“...equality modulo d will below be **denoted** by...”

page 382: The line immediately below equation (20.13) should read:

“...of the interior operator $i_c \equiv i_{\mathbf{X}_c}$...”

page 383: The top displayed equation on this page should read:

$$\psi_2 = \dot{e}^{\mathbf{a}} \wedge e^b \wedge \star R_{ab}.$$

The displayed equation between (20.17) and (20.18) should read:

$$\begin{aligned}\mathfrak{D} \star e^{ab} &= \frac{1}{2} \epsilon^{ab}{}_{cd} \mathfrak{D} e^{cd} = \frac{1}{2} \epsilon^{ab}{}_{cd} (T^c \wedge e^d - e^c \wedge T^d) \\ &= \epsilon^{ab}{}_{cd} T^c \wedge e^d = T_c \wedge \star e^{abc}\end{aligned}$$

page 384: The line of text above equation (20.22) should read:

“In this case the 3-forms $\{\mathcal{G}_c^{(\text{EH})} \equiv -\frac{1}{2}\mathcal{G}_c\}$ define...”

page 385: Equation (20.24) should read:

$$\begin{aligned}\Lambda_{\text{TOTAL}}(e, \omega, A, \phi, \bar{\phi}) &\equiv \frac{1}{\mathbf{2}\kappa} R_{ab}(\omega) \wedge \star e^{ab} - \frac{1}{\mathbf{2}} \kappa_A dA \wedge \star dA \\ &\quad + \frac{1}{\mathbf{2}} \kappa_1 \text{Re}(\overline{\mathfrak{D}_{\text{U}(1)} \phi} \wedge \star \mathfrak{D}_{\text{U}(1)} \phi) \\ &\quad + \frac{1}{\mathbf{2}} \kappa_2 \phi \bar{\phi} \star 1 + \kappa_3 R_{ab}(\omega) \wedge R^{ab}(\omega)\end{aligned}\tag{20.24}$$

The change in minus sign here could be absorbed into the coupling constant κ_A but is changed to be consistent with later sections.

page 386: Equation (20.30) should read:

$$\mathcal{I}_{\text{EH}}[e, \omega] \equiv \frac{1}{2\kappa} \int_{\mathcal{M}} R_{ab}(\omega) \wedge \star(e^a \wedge e^b), \quad (20.30)$$

and the displayed equation below (20.30) becomes:

$$\begin{aligned} \mathcal{G}_c^{(\text{EH})} &\equiv -\frac{1}{2} R_{ab} \wedge i_c \star(e^a \wedge e^b) = 0, \quad a, b, c = 0, 1, 2, 3, \quad e^a(X_c) = \delta_c^a \\ T^c &= 0. \end{aligned}$$

page 388: Equation (20.31) should read:

$$\text{Ein} \equiv (\star^{-1} \mathcal{G}_c^{(\text{EH})}) \otimes e^c = \kappa (\star^{-1} \tau_c^{(\text{SOURCE})}) \otimes e^c. \quad (20.31)$$

page 392: The first equation of section 20.4 should read:

$$\mathcal{I}_{\text{EH}}[e, \omega] \equiv \frac{1}{2\kappa} \int_{\mathcal{M}} R_{ab}(\omega) \wedge \star(e^a \wedge e^b), \quad (20.30)$$

and the second equation becomes:

$$\begin{aligned} \mathcal{I}_{\text{SOURCE}}[e, A, \phi, \bar{\phi}] &\equiv -\frac{1}{2} \kappa_A \int_{\mathcal{M}} \Lambda_2(e, A) \\ &\quad + \frac{1}{2} \kappa_1 \left(\int_{\mathcal{M}} \Lambda_1(e, A, \phi, \bar{\phi}) + \mu^2 \int_{\mathcal{M}} \Lambda_3(e, \phi, \bar{\phi}) \right) \quad (\kappa_1, \mu \in \mathbb{R}) \end{aligned}$$

page 393: The third displayed equation on this page should read:

$$\hat{\delta} \Lambda_3(e, \phi, \bar{\phi}) = \text{Re} \left(\dot{\phi} \wedge \star \bar{\phi} + \phi \wedge \star \dot{\bar{\phi}} + \dot{e}^c \wedge \left[\phi \wedge i_c \star \bar{\phi} - i_c \bar{\phi} \wedge \star \phi \right] \right).$$

The fourth displayed equation should be:

$$\hat{\delta} \Lambda_3(e, \phi, \bar{\phi}) = \text{Re} \left(\dot{\phi} \bar{\phi} + \dot{\bar{\phi}} \phi \right) \star 1 + \dot{e}^c \wedge \left[\phi \bar{\phi} \star e_c \right],$$

the fifth displayed equation should read:

$$\begin{aligned} \hat{\delta} \Lambda_2(e, A) &= d\hat{\delta} A \wedge \star dA + dA \wedge \star \hat{\delta} dA + \dot{e}^c \wedge [dA \wedge i_c \star dA - i_c dA \wedge \star dA] \\ &= d\dot{A} \wedge \star F + F \wedge \star d\dot{A} + \dot{e}^c \wedge [F \wedge i_c \star F - i_c F \wedge \star F] \end{aligned}$$

and the last displayed equation on the page should be:

$$\hat{\delta} \Lambda_2(e, A) \simeq 2 \dot{A} \wedge d \star F + \dot{e}^c \wedge [F \wedge i_c \star F - i_c F \wedge \star F].$$

page 394: The first displayed equation on this page should read:

$$\begin{aligned} \widehat{\delta}\Lambda_1(e, A, \phi, \bar{\phi}) = \text{Re} \Big(& \widehat{\delta}\mathfrak{D}_{U(1)}\bar{\phi} \wedge \star\mathfrak{D}_{U(1)}\phi + \widehat{\delta}\mathfrak{D}_{U(1)}\phi \wedge \star\mathfrak{D}_{U(1)}\bar{\phi} \\ & + \dot{e}^c \wedge [-\mathfrak{D}_{U(1)}\bar{\phi} \wedge i_c \star \mathfrak{D}_{U(1)}\phi - i_c \mathfrak{D}_{U(1)}\phi \wedge \star\mathfrak{D}_{U(1)}\bar{\phi}] \Big) . \end{aligned}$$

The second displayed equation on this page should read:

$$\begin{aligned} \widehat{\delta}\mathfrak{D}_{U(1)}\phi &= d\dot{\phi} + i\hat{s} A \dot{\phi} + i\hat{s} \dot{A} \phi = \mathfrak{D}_{U(1)}\dot{\phi} + i\hat{s} \dot{A} \phi \\ \widehat{\delta}\mathfrak{D}_{U(1)}\bar{\phi} &= d\dot{\bar{\phi}} - i\hat{s} A \dot{\bar{\phi}} - i\hat{s} \dot{A} \bar{\phi} = \overline{\mathfrak{D}_{U(1)}}\dot{\bar{\phi}} - i\hat{s} \dot{A} \bar{\phi} . \end{aligned}$$

The last equation on the page should be:

$$\begin{aligned} \widehat{\delta}\Lambda_1(e, A, \phi, \bar{\phi}) = \text{Re} \Big\{ & \mathfrak{D}_{U(1)}\dot{\bar{\phi}} \wedge \star\mathfrak{D}_{U(1)}\phi + \mathfrak{D}_{U(1)}\dot{\phi} \wedge \star\mathfrak{D}_{U(1)}\bar{\phi} \Big\} \\ & - 2\hat{s} \dot{A} \wedge \text{Im}(\phi \wedge \star\overline{\mathfrak{D}_{U(1)}\phi}) \\ & - \dot{e}^c \wedge \text{Re}(\mathfrak{D}_{U(1)}\phi \wedge i_c \star \mathfrak{D}_{U(1)}\bar{\phi} - i_c \mathfrak{D}_{U(1)}\bar{\phi} \wedge \star\mathfrak{D}_{U(1)}\phi) . \end{aligned}$$

page 395: The fourth displayed equation on this page should read:

$$\begin{aligned} & \text{Re} \Big\{ \mathfrak{D}_{U(1)}\dot{\bar{\phi}} \wedge \star\overline{\mathfrak{D}_{U(1)}\phi} + \mathfrak{D}_{U(1)}\dot{\phi} \wedge \star\mathfrak{D}_{U(1)}\bar{\phi} \Big\} \\ & \simeq \text{Re} \Big\{ -\dot{\phi} \wedge \mathfrak{D}_{U(1)} \star \mathfrak{D}_{U(1)}\bar{\phi} - \dot{\bar{\phi}} \wedge \mathfrak{D}_{U(1)} \star \mathfrak{D}_{U(1)}\phi \Big\} . \end{aligned}$$

The fifth displayed equation on this page should read:

$$\begin{aligned} \widehat{\delta}\Lambda_1(e, A, \phi, \bar{\phi}) &\simeq \text{Re} \Big\{ -\dot{\phi} \wedge \mathfrak{D}_{U(1)} \star \mathfrak{D}_{U(1)}\bar{\phi} - \dot{\bar{\phi}} \wedge \mathfrak{D}_{U(1)} \star \mathfrak{D}_{U(1)}\phi \Big\} \\ & - 2\hat{s} \dot{A} \wedge \text{Im}(\phi \wedge \star\mathfrak{D}_{U(1)}\bar{\phi}) \\ & - \dot{e}^c \wedge \text{Re}(\mathfrak{D}_{U(1)}\phi \wedge i_c \star \overline{\mathfrak{D}_{U(1)}\phi} - i_c \overline{\mathfrak{D}_{U(1)}\phi} \wedge \star\mathfrak{D}_{U(1)}\phi) . \end{aligned}$$

The displayed equation immediately above equation (20.37) should read:

$$\widehat{\delta} \Big(\mathcal{I}_{\text{EH}}[e, \omega] + \mathcal{I}_{\text{SOURCE}}[e, A, \phi, \bar{\phi}] \Big) \simeq 0 ,$$

page 396: Equation (20.38) should read:

$$\kappa_A d \star F + \kappa_1 \hat{s} \operatorname{Im} (\phi \star \overline{\mathfrak{D}_{U(1)} \phi}) = 0 \quad (0.1)$$

and the equation immediately following it becomes:

$$\mathcal{J} \equiv - \frac{\epsilon_0 \kappa_1 \hat{s}}{\kappa_A} \operatorname{Im} (\phi \star \overline{\mathfrak{D}_{U(1)} \phi}) .$$

Equation (20.39) should read:

$$\frac{\mathcal{G}_c^{(\text{EH})}}{\kappa} = \tau_c \quad (20.39)$$

and the text immediately after this displayed equation becomes:

“where $\mathcal{G}_c^{(\text{EH})} \equiv -\frac{1}{2} R_{ab} \wedge i_c \star e^{ab} \dots$ ”.

The displayed equation after (20.39) should read:

$$\begin{aligned} \tau_c = & \frac{\kappa_A}{2} (i_c \star F \wedge \star F - F \wedge i_c \star F) \\ & - \frac{\kappa_1}{2} \operatorname{Re} \left(\mathfrak{D}_{U(1)} \phi \wedge i_c \star \overline{\mathfrak{D}_{U(1)} \phi} - i_c \overline{\mathfrak{D}_{U(1)} \phi} \wedge \star \mathfrak{D}_{U(1)} \phi - \frac{\mu^2}{2} \phi \overline{\phi} \star e_c \right) \quad (c = 0, 1, 2, 3). \end{aligned}$$

The text in the first paragraph immediately below the last displayed equation of this page should be:

“Note $[\tau_c] = \mathbf{M}[c_0^2]$. Since the $\dot{\omega}_{ab}$ partial variations in $\mathcal{I}_{\text{EH}}[e, \omega]$ and $\mathcal{I}_{\text{SOURCE}}[e, A, \phi, \overline{\phi}]$ yield a set of zero-torsion connection 1-forms (see section above), if these are assumed to be metric-compatible with respect to the metric tensor $g = \eta_{ab} e^a \otimes e^b$ then (20.37), (20.38), (20.20) constitute the necessary conditions for the functional $\mathcal{I}_{\text{EH}}[e, \omega] + \mathcal{I}_{\text{SOURCE}}[e, A, \phi, \overline{\phi}]$ to be extremal and (20.39) yields a *Levi-Civita Einstein tensor field equation with sources*. ”

page 402: The penultimate displayed equation on this page should read:

$$\mathbf{T}^{(\text{MAX})}(X_0, X_0) = (\star \tau_c^{(\text{MAX})})(X_0) \delta_0^c = i_0(\star \tau_0^{(\text{MAX})})$$

page 405: The second displayed equation in section 21.1 should read:

$$(\nabla \cdot \Xi)(\mathbf{Y}) \equiv (\nabla_{X_a} \Xi)(X^a, Y) \quad \text{for all } Y \in \Gamma T\mathcal{U}_{\mathcal{M}}.$$

page 414: Equation (22.8) should read:

$$\mathcal{G}_a^{(\text{EH})} = \kappa \tau_a^{(\text{SOURCE})}, \quad a = 0, 1, 2, 3. \quad (22.8)$$

The text immediately following this line should then read:

“Recall that it follows from $\left[\mathcal{G}_a^{(\text{EH})} \right]$ and . . .”.

Equation (22.9) should read:

$$\mathcal{G}_a^{(\text{EH})} = \kappa \sum_N \tau_a^N \equiv \kappa \tau_a^{(\text{SOURCE})}. \quad (22.9)$$

page 418: The first displayed equation on this page should read:

$$J_{K_j}^{(V)} \equiv -\frac{\textcolor{red}{c}_0}{\textcolor{red}{L}_0^2} i_V \tau_{K_j}^{(\text{MAX})}, \quad [\textcolor{red}{K}_j] = \frac{\textcolor{red}{T}}{\textcolor{red}{L}}, \quad (j = 1, 2, 3).$$

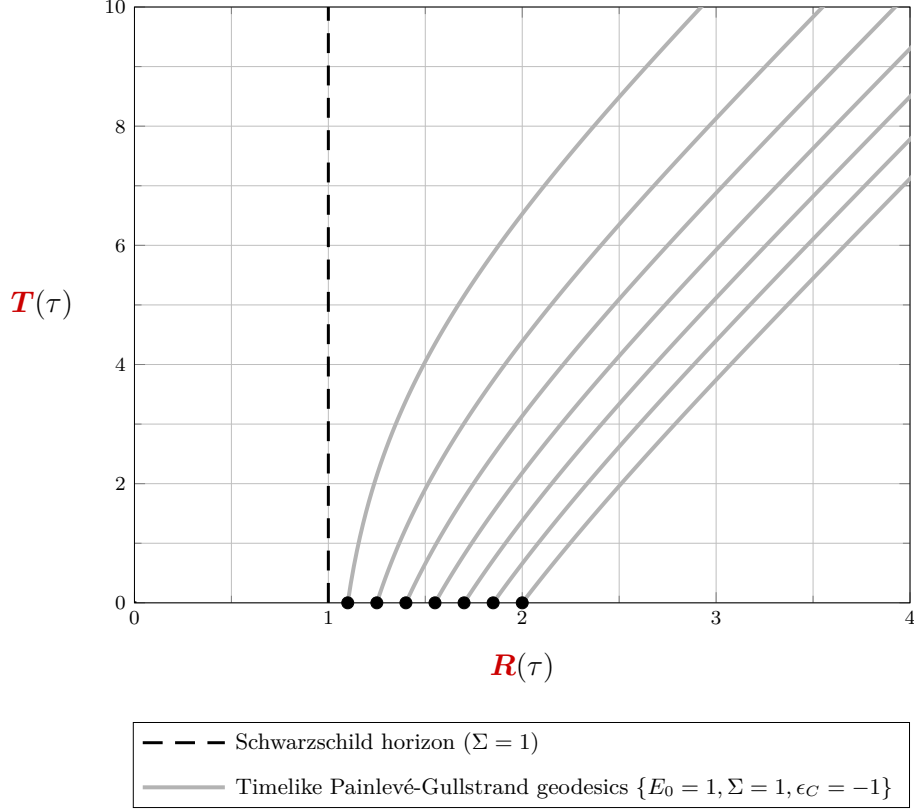
The penultimate displayed equation on this page should read:

$$\begin{aligned} \star(F \wedge \star F) = \frac{L_0^2}{8\pi^2 \epsilon_0^2} \left(\frac{q_2^2}{2r^4} + \frac{q_1 q_2 (r^2 - x^3 d_0)}{\textcolor{red}{r}^3 ((x^1)^2 + (x^2)^2 + (x^3 - d_0)^2)^{3/2}} \right. \\ \left. + \frac{q_1^2}{2((x^1)^2 + (x^2)^2 + (x^3 - d_0)^2)^2} \right) \end{aligned}$$

page 488: The first displayed equation on this page should read:

$$\begin{aligned} \frac{1}{L_0} \mathfrak{g}_{\text{PG}}(\dot{C}, X_0) \Big|_{C(\tau)} &= -\dot{T}(\tau) \\ \frac{1}{L_0^2} \mathfrak{g}_{\text{PG}}(\dot{C}, \dot{C}) \Big|_{C(\tau)} &= - \left(1 - \frac{\Sigma}{R(\tau)} \right) \dot{T}(\tau)^2 + 2 \sqrt{\frac{\Sigma}{R(\tau)}} \dot{T}(\tau) \dot{R}(\tau) + \dot{R}(\tau)^2 \\ &\quad + R(\tau)^2 \left(\dot{\Theta}(\tau)^2 + \sin^2(\textcolor{red}{\Theta}(\tau)) \dot{\Phi}(\tau)^2 \right). \end{aligned}$$

page 491: The axis labels ($T(\tau), R(\tau)$) in Fig. 30.1 are the wrong way round and need to be swapped. The figure should be as follows:



page 573: The penultimate displayed equation on this page should read:

$$f(r^*) \equiv r^* + \frac{(\Sigma_1^2 - 2\Sigma_2^2)}{\sqrt{4\Sigma_2^2 - \Sigma_1^2}} \arctan\left(\frac{2r^* - \Sigma_1}{\sqrt{4\Sigma_2^2 - \Sigma_1^2}}\right) + \frac{\Sigma_1}{2} \ln\left((r^*)^2 - \Sigma_1 r^* + \Sigma_2^2\right).$$

page 574: The first displayed equation on this page has an extra dr^* in the first term that needs removing. It should read:

$$\left\{ \begin{array}{l} \Psi^* dt = d(\Psi^* t) = d(t^* + f(r^*)) = dt^* + \frac{dr^*}{1 - \frac{\Sigma_1}{r^*} + \frac{\Sigma_2^2}{(r^*)^2}} \\ \Psi^* dr = d(\Psi^* r) = dr^* \\ \Psi^* d\theta = d(\Psi^* \theta) = d\theta^* \\ \Psi^* d\phi = d(\Psi^* \phi) = d(\phi^*) = d\phi^* \end{array} \right.$$

The Reissner-Nordström -star metric (30.180) should then read:

$$\frac{\mathfrak{g}_{\text{RN}^*}}{L_0^2} \equiv \frac{\Psi^* \mathfrak{g}_{\text{RN}}^{(t,r,\theta,\phi)}}{L_0^2} = - \left(1 - \frac{\Sigma_1}{r^*} + \frac{\Sigma_2^2}{(r^*)^2} \right) dt^* \otimes dt^* - dt^* \otimes dr^* - dr^* \otimes dt^* \\ + (r^*)^2 (d\theta^* \otimes d\theta^* + \sin^2(\theta^*) d\phi^* \otimes d\phi^*) . \quad (30.180)$$

page 659: Equation (34.14) for the Bopp-Podolsky field equations should read:

$$d \star \mathcal{F} + \lambda_0^2 d\delta d \star \mathcal{F} = 0 \quad (34.14)$$

The change in sign here ensures the static, spherically symmetric solution is finite as $r \rightarrow 0$. This correction contributes to a number of changes over the next few pages.

page 660: Equation (34.16) for the Bopp-Podolsky field equations should read:

$$d \star \mathcal{F} + \lambda_0^2 d\delta d \star \mathcal{F} = d\delta d\alpha \wedge \Pi + \lambda_0^2 d\delta d\delta d\alpha \wedge \Pi \\ = d \left(\delta d\alpha + \lambda_0^2 \delta d\delta d\alpha \right) \wedge \Pi = 0. \quad (34.16)$$

The following equation also needs correcting:

$$\square \alpha + \lambda_0^2 \square^2 \alpha = 0, \quad \square \alpha \neq 0, \quad (34.17)$$

as does the final displayed equation of the page which should read:

$$\square \alpha + \frac{1}{\lambda_0^2} \alpha = 0, \quad \lambda_0 \neq 0.$$
